Autonomous Drivetrain: Control Methods

VEX EDR 4-motor differential drive — open-loop and closed-loop position/range control

1. SYSTEM MODEL

The drivetrain is treated as a one-degree-of-freedom system: a point mass traveling along a straight line from an origin to a target distance.

X — distance traveled so far (derived from encoder ticks, or from range readings for station-keeping)
Xd — desired (target) distance
e = Xd − X — position error
α — maximum commanded speed

Three control strategies were implemented against this model, increasing in feedback sophistication, plus one variant that reuses the proportional law against a different sensor.

2. METHOD 1 — TIME-BASED CONTROL (OPEN-LOOP)

No position feedback. Wheel speed at a given motor power is characterized empirically off-robot; the required run time is then computed from the target distance and that speed estimate, and the motors are driven blind for that duration.

$$t_{run} = 2 \cdot X_d / (\omega \cdot d_{wheel})$$

where ω is the measured wheel angular velocity at the commanded power level, and d_{wheel} is the wheel diameter.

Accuracy is entirely dependent on the speed model holding true on the day — wheel slip, surface friction, and battery voltage are all unmodeled and uncorrected.

> Implementation: `01_open_loop_timed_drive.c`

3. METHOD 2 — ON-OFF CONTROL (CLOSED-LOOP)

A quadrature encoder is added to measure actual wheel rotation. The target distance is converted to a target encoder tick count; motors run at fixed full power until that count is reached, then power is cut immediately.

This removes the dependency on the open-loop speed model — distance traveled is now measured, not estimated. However, the controller is strictly on/off: there is no deceleration profile, so the robot decelerates abruptly at the target, which is more likely to overshoot due to momentum.

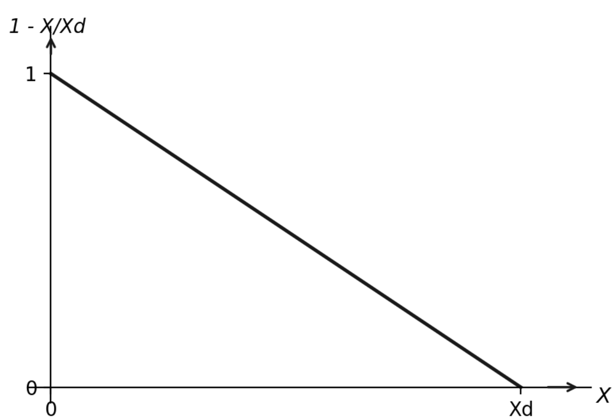
> Implementation: `02_closed_loop_bangbang_drive.c`

4. METHOD 3 — PROPORTIONAL (P) CONTROL ON POSITION

Commanded speed is scaled continuously with remaining distance, rather than switched on/off, producing a linear deceleration profile into the target.

$$\dot{X} = \alpha \left(1 - \frac{X}{X_d} \right) = \alpha \left(\frac{X_d - X}{X_d} \right) = \bar{\alpha} \cdot e$$

where $e = X_d - X$ (position error), $\bar{\alpha} = \alpha / X_d$ (effective proportional gain), valid for $0 < X < X_d$



The commanded speed is maximal (α) at the start of the run and decays linearly to zero as X approaches X_d. This is a proportional controller with gain $\bar{\alpha}$ acting on position error e.

> Implementation: `03_closed_loop_proportional_drive.c`

5. METHOD 4 — PROPORTIONAL CONTROL ON RANGE (STATION-KEEPING)

The Method 3 control law is reapplied against a different feedback signal and a different objective: instead of a one-shot point-to-point move, the controller continuously holds the robot at a fixed standoff distance from an object, using ultrasonic range in place of encoder position.

$$speed = K_p \cdot e\,, \quad e = R - R_d$$

where R is the measured sonar range, R_d is the desired standoff distance (50 in.), and K_p = 2 in the current implementation. The loop runs continuously at 10 Hz rather than terminating once a target is reached.

> Implementation: `04_closed_loop_sonar_station_keeping.c`

Known limitation: the current implementation commands `abs(speed)`, discarding the sign of e. The controller can therefore only drive forward (faster or slower); it cannot reverse to back away from an object closer than R_d. A correct bidirectional station-keeping controller would preserve the sign of e.

SUMMARY

Method	Feedback	Control Law	Behavior at Target
1. Time-based	None (open-loop)	Fixed power, fixed duration	Stops when time elapses
2. On-off	Encoder (position)	Fixed power until X = X_d	Hard stop
3. Proportional	Encoder (position)	Speed ∝ (X_d − X)	Smooth deceleration
4. Proportional (range)	Sonar (range)	Speed ∝ (R − R_d), continuous	Holds standoff distance